

Chemistry 3A

Introductory General Chemistry

Math of Chemistry: Scientific Notation,
Significant Digits/Figures, Conversions,
Density, Temperature Scales

- Math in review
- (Measured) Quantities
- Scientific Notation / Standard Notation
- Precision: Significant Digits (Significant Figures)
- Decimal Places
- Operations on Numbers (multiplication/division, addition/subtraction) & How Precision Affected
- Conversions / Dimensional Analysis
- Density
- Temperature: Scales, Conversion

Math in Chem: Algebra Review

- The quantity a
- The inverse of the quantity: $\frac{1}{a} = 1/a$
- The inverse of the inverse takes you back: $\frac{1}{\left(\frac{1}{a}\right)} = a$

$$\frac{\left(\frac{a+b}{c-d}\right)}{\left(\frac{e+f}{g-h}\right)} = \frac{(a+b)(g-h)}{(c-d)(e+f)}$$

- “Implied denominator of 1”. It might help when solving chemistry problems

$$\text{The number } 2.4 = \frac{2.4}{1}$$

Quantities

Number	Unit	Dimension
1	mile	length (distance)
0.25	liters	volume
15	amperes	electric current

A (measured) **quantity** in science, in chemistry has a **number** and a **unit** associated with it

- The **number** is how many/much
- The **unit** is a scale of the measurement, the how many/much of what

Scientific Notation

A way of expressing numerical values

Format: $d.mmm \times 10^n$

<i>Part</i>	$d.mmm$
<i>Part Name</i>	Significand (preferred), Coefficient, (Mantissa is obsolete term)
<i>Part Property</i>	Number greater than or equal to 1.0 and less than 10. This number properly express significant digits
<i>Part</i>	10^n
<i>Part Name</i>	Exponent
<i>Part Property</i>	Exponent of 10 where n is integer, negative, positive (note $10^0 = 1$)

Scientific Notation

decimal point moves to right

$$1. = 10^0$$

$$10. = 10^1$$

$$100. = 10^2$$

decimal point moves to left

$$0.1 = 10^{-1} = 1/10 = \frac{1}{10} \text{ (expressed in various ways)}$$

$$0.01 = 10^{-2} = 1/100 = \frac{1}{100}$$

$$0.001 = 10^{-3} = 1/1000 = \frac{1}{1000}$$

Scientific Notation

1. Is the number less than 10 and greater than 1?

You don't really need scientific notation, because it will be
[$d.mmm \times 10^0$] and $10^0 = 1$, so $\rightarrow d.mmm$

2. Otherwise take the number and express as $d.mmm$

3. If original number is greater than $d.mmm$
you have to multiply by 10^1 n times to get to
original number, so n will be positive (greater
than zero): $n \times 1 = n$ (greater than zero)

4. If original number less than $d.mmm$
you have to multiply by 10^{-1} n times to get to
original number, so n will be positive (greater
than zero): $n \times -1 = -n$ (less than zero)

Scientific Notation

Reiterating #3

3. If original number is greater than *d.mmm* you have to multiply by $10^{\textcolor{red}{1} \textcolor{brown}{n}}$ times to get to original number, so *n* will be positive (greater than zero): $\textcolor{brown}{n} \times \textcolor{red}{1} = +\textcolor{brown}{n}$

Your book: "if you moved decimal point to the left *n* places, then *n* is positive"

$35 \rightarrow 35. \rightarrow 3.5$ *moved to left 1 place*

$$3.5 \times 10^1$$

$23849 \rightarrow 23849. \rightarrow 2.3849$ *moved to left 4 places*

$$2.3849 \times 10^4$$

Scientific Notation

Reiterating #4

4. If original number less than $d.mmm$ you have to multiply by 10^{-1} (or divide by 10) n times to get to original number, so n will be negative (less than zero): $n \times -1 = -n$

Your book: "if you moved decimal point to the right n places, then n is negative"

$0.0000035 \rightarrow 3.5$ *moved to right 6 places*

$$3.5 \times 10^{-6}$$

$0.023849 \rightarrow 2.3849$ *moved to right 2 places*

$$2.3849 \times 10^{-2}$$

Scientific Notation

Patterns for memorizing

- *decimal point movement – number being changed – final exponent value*
- Left – Large – Positive
- Right – Small – Negative

Positive and negative for **exponent**, not **significand**!

$$34249 \rightarrow 3.4249 \times 10^4$$

$$-34249 \rightarrow -3.4249 \times 10^4$$

$$0.0034249 \rightarrow 3.4249 \times 10^{-3}$$

$$-0.0034249 \rightarrow -3.4249 \times 10^{-3}$$

Scientific Notation

Scientific notation is *actually* a *product* of *two numbers*: the *significand* and a *power of 10*!

$$34249 = 3.4249 \times 10^4 = 3.4249 \times 10,000 = 34249$$

$$-34249 = -3.4249 \times 10^4 = -3.4249 \times 10,000 = -34249$$

$$0.0034249 = 3.4249 \times 10^{-3} = 3.4249 \times 0.001 = 0.0034249$$

$$-0.0034249 = -3.4249 \times 10^{-3} = -3.4249 \times 0.001 = -0.0034249$$

Use this feature to confirm whether you converted a number to scientific notation properly!

Practicing Scientific Notation

Let's start with an "original" number
one not in scientific notation

This number will be (much) greater than 10

238,700

Two hundred thirty-eight thousand seven hundred

Practicing Scientific Notation

238,700

Our goal will be to get to a number between 1 and 10 since scientific notation requires it, and it should include the leftmost non-zero digit to the rightmost non-zero digit

This span of digits will be the **significand**

The significand should in the end be of *d.mmm...* form 2.387

So we need to determine the **exponent** (power of 10)

Practicing Scientific Notation

1. Place the “implied” decimal point where it should be

238,700.

2. Move it in the direction of where it should end up

This is to the left to produce a 2.387 significand

23,870.

3. What is the relationship between 238,700 and 23,870? You have to multiply by 10!

That is $238,700 = 23,870 \times 10$

We also note that $10 = 10^1$

So $238,700 = 23,870 \times 10^1$

Let's repeat the movement until we get to proper form

Practicing Scientific Notation

238,700. (the original number)

$$23,870. \rightarrow 23,870 \times 10^1 = 238,700$$

$$2,387. \rightarrow 2,387 \times 10^2 = 238,700$$

$$238.7 \rightarrow 238.7 \times 10^3 = 238,700$$

Note that we were dropping zeroes until we hit the non-zero number "7", but the decimal point still moves

$$23.87 \rightarrow 23.87 \times 10^4 = 238,700$$

$$2.387 \rightarrow 2.387 \times 10^5 = 238,700$$

The decimal point was not moved to create the significand: we have a number less than 10, greater than 1. With each movement of decimal point, we created the power of 10. At every step, equal to original number

Practicing Scientific Notation

Let's try a number much less than 1

0.00003049 (the original number)

Following the rules/format, the **significant** should look like 3.049 in the end

$$0.0003049 / 10 = 0.00003049$$

$$0.0003049 \times 10^{-1} = 0.00003049$$

$$0.003049 / 100 = 0.00003049$$

$$0.003049 \times 10^{-2} = 0.00003049$$

$$0.03049 / 1000 = 0.00003049$$

$$0.03049 \times 10^{-3} = 0.00003049$$

$$0.3049 / 10000 = 0.00003049$$

$$0.3049 \times 10^{-4} = 0.00003049$$

$$3.049 / 100000 = 0.00003049$$

$$3.049 \times 10^{-5} = 0.00003049$$

Significant Digits

Significant digits indicate the precision, the confidence of a single measurement

Also called significant figures

Measurement /Value	Significant Digits	Explanation
1.7	2	All non-zero digits significant
1.732	4	
17.32704	7	Zeroes between non-zero digits are significant

Significant Digits

Measurement /Value	Significant Digits	Explanation
0.0045	2	leading zeroes after decimal point not significant
0.00450	3	trailing zeroes after decimal are significant
1200	2	
1200.	4	assertion of decimal point makes significant
0.000	0	no non-zero digits at all
1000.0	5	adding decimal point and any zeroes after it makes all digits significant
100	1	evaluated like the “1200” example above
1.200×10^{-3}	4	all digits in scientific notation should be significant

Significant Digits

Significant digits indicate the precision, the confidence of a single measurement

Preparative balance

- Measuring quantities usually 0.1 g (100 mg) or greater
- used to weigh masses for starting a chemistry experiment
- Precision example: 3.4 g sodium chloride (NaCl)



Analytical balance

- Measuring quantities usually down to 0.0001 g (0.1 mg or 100 μ g) or greater
- used to weigh masses for determining yield of product at end of a chemistry experiment
- Precision example: 0.0241 g copper(II) sulfate pentahydrate



$$18 \neq 18. \neq 18.0 \neq 18.00$$

- They are all mathematically equal to 18
- But they are not the same in terms of the precision or certainty of the value

18 → 2 significant digits: ± 1

18. → also 2 significant digits: ± 1

But the measuring device will show decimals

18.0 → 3 significant digits: ± 0.1

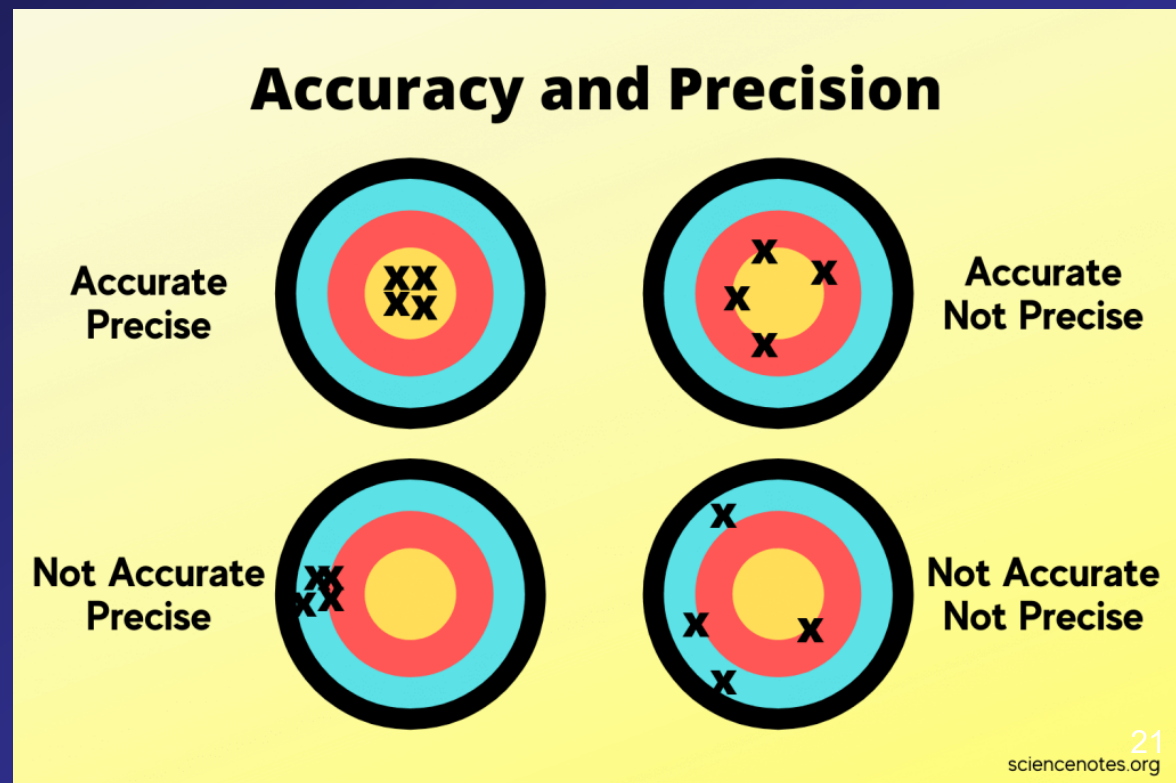
18.00 → 4 significant digits: ± 0.01

A Slight Tangent on “Precision”

Multiple Measurements

Precision – an indicator of how measurements **repeat** the value being measured

Accuracy – an indicator of how measurements get to the **true** value



Significant Digits in Scientific Notation

The rules on significant digits in numerical values always apply in scientific notation

$$2.69 \times 10^{-5} = 0.0000\textcolor{teal}{269} \quad (3 \text{ significant digits})$$

$$2.69000 \times 10^{-5} = 0.0000\textcolor{teal}{269000} \quad (6 \text{ significant digits})$$

$$2.69 \times 10^3 = \textcolor{teal}{2690} \quad (3 \text{ significant digits})$$

You still have add the zero to get standard form of number as integer

$$2.69000 \times 10^3 = \textcolor{teal}{2690.00} \quad (6 \text{ significant digits})$$

Doing Math with “Decimal Places”

Addition and Subtraction of Numbers with Different Decimal Places

Decimal places for final answer should be the one number with the fewest number of decimal places

Solve $12.11 + 18.0 + 1.013 =$

				Result	
				Calculator's Display	Your Answer
Value	12.11	18.0	1.013	31.123	31.1
Decimal Places	2	1	3	3	1

Solve $100.29 - 2.343 - 72.9 =$

				Result	
				Calculator's Display	Your Answer
Value	100.29	2.343	72.9	25.047	25.0
Decimal Places	2	3	1	3	1

Solve $1000.23 - 87.532 + 200.49 - 50.439 =$

					Result	
					Calculator's Display	Your Answer
Value	1000.23	87.532	200.49	50.439	1062.749	1062.75
Decimal Places	2	3	2	3	3	2

Doing Math with Sig Digits

Multiplication and Division of Numbers with Different Significant Digits

Significant digits for final answer should be the one number with the fewest number of significant digits

Solve $4.56 \times 1.4 =$

			Result	
			Calculator's Display	Your Answer
Value	4.56	1.4	6.384	6.4
Sig Digits	3	2	-	2

Solve $834.4 \div 34.92 \times 2.104 =$

				Result	
				Calculator's Display	Your Answer
Value	834.4	34.92	2.104	50.27427262	50.27
Sig Digits	4	4	4	-	4

Solve $2340.09 \div 24.190 =$

			Result	
			Calculator's Display	Your Answer
Value	2340.09	24.190	50.27427262	50.27
Sig Digits	6	5	-	4

Rules for Significant Figures and Precision in Chemistry Calculations

1. Addition and Subtraction

When adding or subtracting, the final answer is rounded to match the number of **decimal places** of the number in the calculation with the **fewest decimal places**.

Example:

$100.29 - 2.343 - 72.9 = 25.047 \rightarrow$ Rounded to 1 decimal place $\rightarrow$ **25.0**

2. Multiplication and Division

When multiplying or dividing, the final answer is rounded to have the same number of **significant figures** as the measurement with the **fewest significant figures**.

Example:

$4.56 \times 1.4 = 6.384 \rightarrow$ 2 significant figures $\rightarrow$ **6.4**

3. Rounding Advice

Always perform all intermediate calculations first (keep extra digits in your calculator). Round only at the very end to avoid introducing rounding errors.

Tip: Underline the number in your intermediate result that determines your rounding.

Operation	Rule
Addition / Subtraction	Match decimal places to least precise measurement
Multiplication / Division	Match significant figures to least precise measurement

Doing Calculations: NO INTERMEDIATE ROUNDING!!!

Simple example

Calculate the cost of carpeting a 4.8 m x 3.2 m floor where the carpeting costs \$42 per square meter

Suppose you do the calculation in steps (intermediate calculations):

1. The area: $4.8 \text{ m} \times 3.2 \text{ m} = 15.36 \text{ m}^2$ (value on calculator display)

2. Finishing off

rounding no decimal: $15 \text{ m}^2 \times (\$42 / \text{m}^2) = \630.00

rounding to one decimal: $15.4 \text{ m}^2 \times (\$42 / \text{m}^2) = \$646.80$

no rounding: $15.36 \text{ m}^2 \times (\$42 / \text{m}^2) = \$645.12$

No need for intermediate result determination in (nearly) all cases:

$4.8 \text{ m} \times 3.2 \text{ m} \times (\$42 / \text{m}^2) = \$645.12$

In business accounting & chemistry, do the rounding at the end of the calculation. To significant digits (or decimal places)

Doing Chem Math

- Addition and Subtraction
 - look at the DECIMAL PLACES!
- Multiplication and Division
 - look at the SIGNIFICANT DIGITS!

The Book

1. Do the Multiplication and Division first
2. Then do Addition and Subtraction operations

If you do intermediate rounding—I wouldn't!—then if you follow the book, this is acceptable

PEMDAS

PEMDAS -- (P)arentheses, (E)xponents, (MD)
Multiplication/Division, (AS) Addition/Subtraction

$$5 + 6 \times 2 - 8 / 4$$

- P and E: none
- MD: $5 + 12 - 2$
- AS: 15

$$14^2 + 633 / 3 - 14 \times 8$$

- P: none
- E: $196 + 633 / 3 - 14 \times 8$ (did 14^2)
- MD: $196 + 211 - 112$
- AS: 295

PEMDAS

PEMDAS -- (P)arentheses, (E)xponents, (MD)
Multiplication/Division, (AS) Addition/Subtraction

$$(22.17 + 100.24) \times (443.4 + 349.3) / 4^3$$

- P: $122.41 \times 792.7 / 4^3$
- E: $122.41 \times 792.7 / 64$ (64 is not a MEASUREMENT!)
- MD: 1516.162609
- AS: 1516.162609 ==> **1516**

PEMDAS

PEMDAS -- (P)arentheses, (E)xponents, (MD)
Multiplication/Division, (AS) Addition/Subtraction

$$100.13 - 23.433 / ((443.4 + 349.3) / 2^4)$$

- P: $100.13 - 23.433 / (792.7 / 2^4) \rightarrow$
 $100.13 - 23.433 / (792.7 / 16) \rightarrow$
 $100.13 - 23.433 / 49.54375$
- E: evaluated within parentheses!
- MD: $100.13 - 0.47297590513$
- AS: $99.6570240949 ==> \mathbf{99.66}$

Caution In Using Your Calculator

- In your lab, you had the following masses:

Item	Mass
Empty beaker	25.4833 g
Beaker with sodium chloride (NaCl) reagent	29.8573 g
(calculation) Sodium chloride (NaCl) used*	4.3740 g
* Difference in mass = $29.8573 - 25.4833 = 4.374$	

- You had to get two masses on the balance, then make a calculation to determine the difference to get the mass of a chemical
- Your calculator gave a result "4.374"
- But rules on precision inform you it is "4.3740", not "4.374"
- So, you need to be aware of precision when your calculator is not

Conversion Factors

Calculations in chemistry frequently require converting between the **units** of a **quantity**

- Volume: fluid ounces to milliliter ("fl oz" to "ml")
- Temperature: Celsius to Kelvin ("°C" to "K")
- Pressure: torr to atmosphere ("torr" to "atm")

Conversions not just about measurement systems but also include scales which are powers of 10

- 1 liter (L) = 1000 milliliters = 1000 mL
not 1000 ml! use SI unit lettering case
- 1 kilogram (kg) = 1000 grams (g)

Conversion Nomenclature

Prefix	Meaning	Example
<i>Multiple of 10 (greater than 1)</i>		
kilo- (k)	$10^3 = 1000$	2 kilograms (kg) = 2000 grams
mega- (M)	$10^6 = 1,000,000$	3.7 megajoules (MJ) = 3,700,000 J
giga- (G)	$10^9 = 1,000,000,000$	10 gigahertz (GHz) = 10,000,000,000 Hz
<i>Submultiples of 10 (fractions of 1)</i>		
deci- (d)	$10^{-1} = 1/10$	1 deciliter (dL, dl) = 0.1 L
centi- (c)	$10^{-2} = 1/100$	2.54 centimeter (cm) = 0.0254 m
milli- (m)	$10^{-3} = 1/1000$	14 millibar (mbar) = 0.014 bar
micro- (u/μ)	$10^{-6} = 1/1000000$	259 microliters (μl, ul, μL, uL) = 0.000259 L
nano- (n)	$10^{-9} = 1/1000000000$	34 nanograms (ng) = 0.000000034 g
pico- (p)	$10^{-12} = 1/1000000000000$	250 picoliters (pL) = 0.00000000025 L
femto- (f)	$10^{-15} = 1/1000000000000000$	4.3 femtomoles (fmol) = 0.000000000000043 mol

units in green are often used in chemistry

Conversion Practice

A room measures 16 ft x 20 ft in area. What is it in square meters (m²)

A conversion problem is going to be an equation usually involving a mathematical product (the multiplication of quantities), and these quantities will include your given quantities and conversion factors

$$16 \text{ ft} \times 12 \text{ ft} \times \dots = ? \text{ m}^2$$

Conversion Practice

A room measures 16 ft x 20 ft in area. What is it in square meters (m²)

I remember

- 12 inches = 1 foot
- 2.54 cm = 1 inch
- 100 cm = 1 meter

These are actually conversion factors !

We have to get enough conversion factors to go from units on the left to the result with unit on right of equation!

$$16 \text{ ft} \times 12 \text{ ft} \times \dots = ? \text{ m}^2$$

Conversion Practice

A room measures 16 ft x 20 ft in area. What is it in square meters (m^2)

- 12 inches = 1 foot
- 2.54 cm = 1 inch
- 100 cm = 1 meter

We know we have to multiply 16 ft x 12 ft to get an area → that will give us a value in square feet (ft^2)

$$16 \text{ ft} \times 12 \text{ ft} \times \dots = ? \text{ m}^2$$

Conversion Practice

A room measures 16 ft x 20 ft in area. What is it in square meters (m²)

- 12 inches = 1 foot
- 2.54 cm = 1 inch
- 100 cm = 1 meter

So we need a conversion factor that is in ft²

$$12 \text{ in} = 1 \text{ ft} \rightarrow (12 \text{ in})^2 = (1 \text{ ft})^2$$

I am not going to evaluate it right now: I will add it to the equation

$$\frac{16 \text{ ft}}{1} \times \frac{12 \text{ ft}}{1} \times \frac{(12 \text{ in})^2}{(1 \text{ ft})^2} \times \dots = ? \text{ m}^2$$

Conversion Practice

$$\frac{16 \text{ ft}}{1} \times \frac{12 \text{ ft}}{1} \times \frac{(12 \text{ in})^2}{(1 \text{ ft})^2} \times \dots = ? \text{ m}^2$$

I added the conversion factor in a way that when "ft x ft" is evaluated, it equals "ft²"

If we evaluate that expression right now and stop, we drop the units to get just the numbers, and we get an answer of $16 \times 12 \times 12^2 = 27468$

The $\text{ft} \times \text{ft} / \text{ft}^2$ should "cancel": two "ft" in the numerator, "ft²" in the denominator "in²" is left in the numerator, so that would be the intermediate result: "27,468 in²"

Conversion Practice

$$\frac{16 \text{ ft}}{1} \times \frac{12 \text{ ft}}{1} \times \frac{(12 \text{ in})^2}{(1 \text{ ft})^2} \times \frac{(2.54 \text{ cm})^2}{(1 \text{ in})^2} \times \frac{(1 \text{ m})^2}{(100 \text{ cm})^2} = ? \text{ m}^2$$

Note that we had to use the squared version of these linear conversions to get cancellation of the units in the quantities, but this is to demonstrate what you may sometimes need to keep track of. The book discusses this too.

When you have ensured all the units cancel, remove them from the equation, and evaluate the expression on the left

$$\frac{16}{1} \times \frac{12}{1} \times \frac{(12)^2}{(1)^2} \times \frac{(2.54)^2}{(1)^2} \times \frac{1^2}{100^2} = 17.8 \text{ m}^2$$

Conversion Practice

- Now I suppose you could look up in a table that $1 \text{ ft}^2 = 0.092903 \text{ m}^2$

$$\frac{16 \text{ ft}}{1} \times \frac{12 \text{ ft}}{1} \times \frac{0.092903 \text{ m}^2}{1 \text{ ft}^2} = 17.8 \text{ m}^2$$

Just keep in mind what the conversion factor is and how you use it to cancel **units** properly in evaluating the product. Ensure that the resulting final **unit** is what it should be.

Density

How much **mass** is in a given **volume**

The math definition:

Greek letter *rho* (ρ) is
symbol for **density**

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

$$\rho = \frac{m}{V}$$

Densities of Natural Substances

Table 1.4.1: Densities of Common Substances

Substance	Density at 25°C (g/cm ³)
blood	1.035
body fat	0.918
whole milk	1.030
corn oil	0.922
mayonnaise	0.910
honey	1.420



Problem Solving Process Using Density

The problem/question (from your book)

A mercury thermometer for measuring a patient's temperature contains 0.750 g of mercury.

What is the volume of this mass of mercury?

The most important thing you will ever do is to spend the time and effort to fully understand the question, and identify the key elements of it

Look for what is asked for: volume of mercury

Look for what is given: a mass (0.750 g) of mercury

Problem Solving Process Using Density

What is missing from the information given in the problem/question?

Usually the information not given are constants of nature and (numerical) properties of matter and energy like atomic weights.

In this case, it is a density.

The density of a substance is a property that allows you to relate its mass to its volume *and vice versa*

The density of water is 1 g/mL (at 0°C) and is a property of water

The density of mercury is 13.6 g/mL

Problem Solving Process Using Density

For this equation about density, this is all just algebra below which you already know

$$\rho = \frac{m}{V} \rightarrow V \times \rho = m \rightarrow V = \frac{m}{\rho} = \frac{1}{\rho} \times m$$

In the last manipulation, we have an expression that tells us what the volume is equal to: the inverse of the density multiplied by the mass, or “mass over density”

Problem Solving Process Using Density

$$\frac{1}{\rho} \times m = V$$

We now just enter the values to solve:

$$\frac{1}{13.6 \text{ g/mL}} \times 7.50 \text{ g} = \frac{1 \text{ mL}}{13.6 \text{ g}} \times 7.50 \text{ g} = 0.55 \text{ mL}$$

Could be expressed as $5.5 \times 10^{-1} \text{ mL}$

Or converting to microliters (μL)

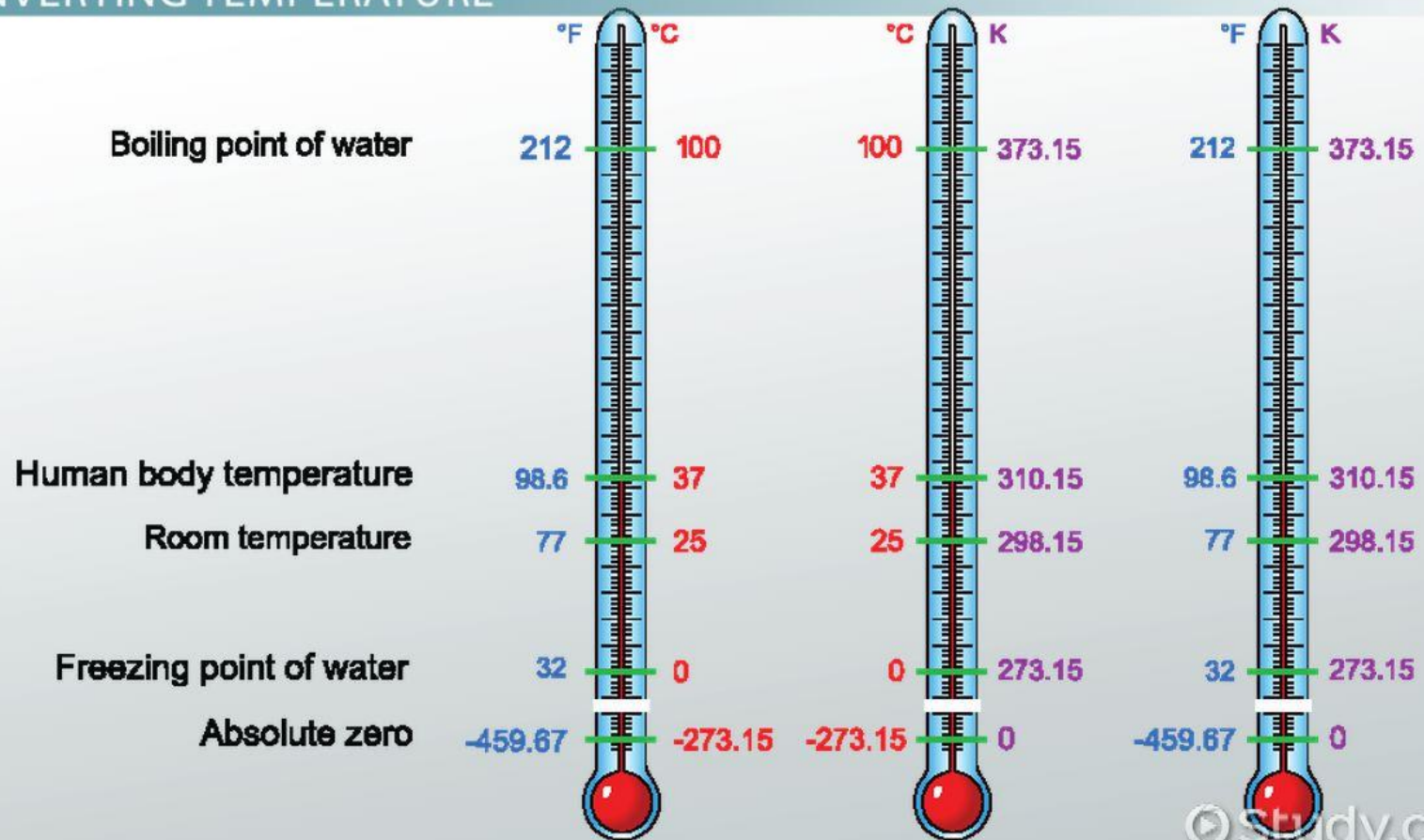
$$0.55 \text{ mL} \times \frac{1000 \mu\text{l}}{1 \text{ mL}} = 550 \mu\text{L}$$

Temperature

- A measure of the average kinetic energy of particles in a substance
- Not energy itself, but an indicator or measure of energy

Measuring Temperature

CONVERTING TEMPERATURE



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Temperature Scales

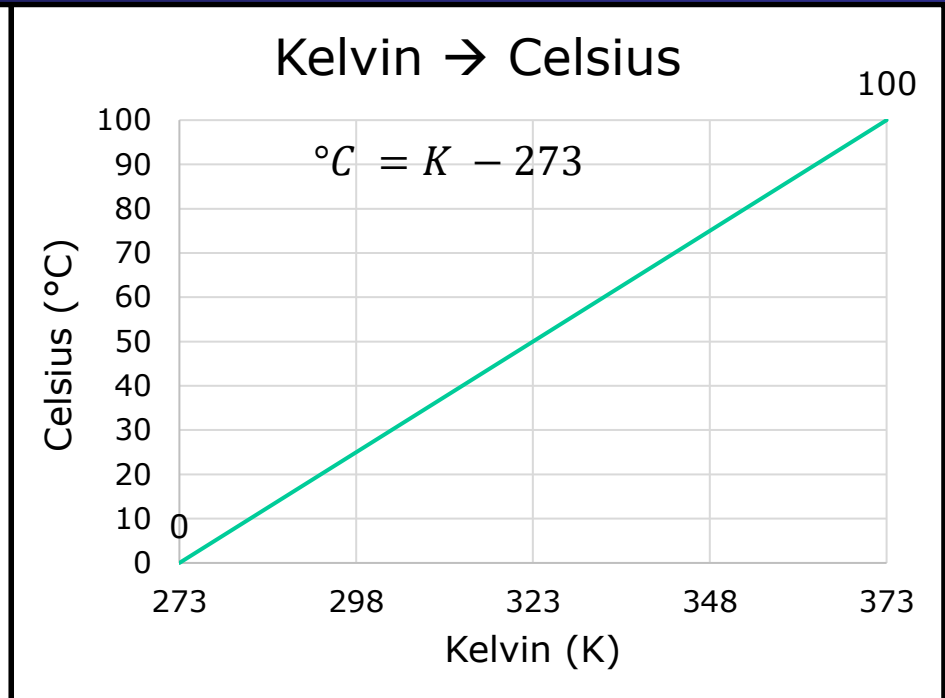
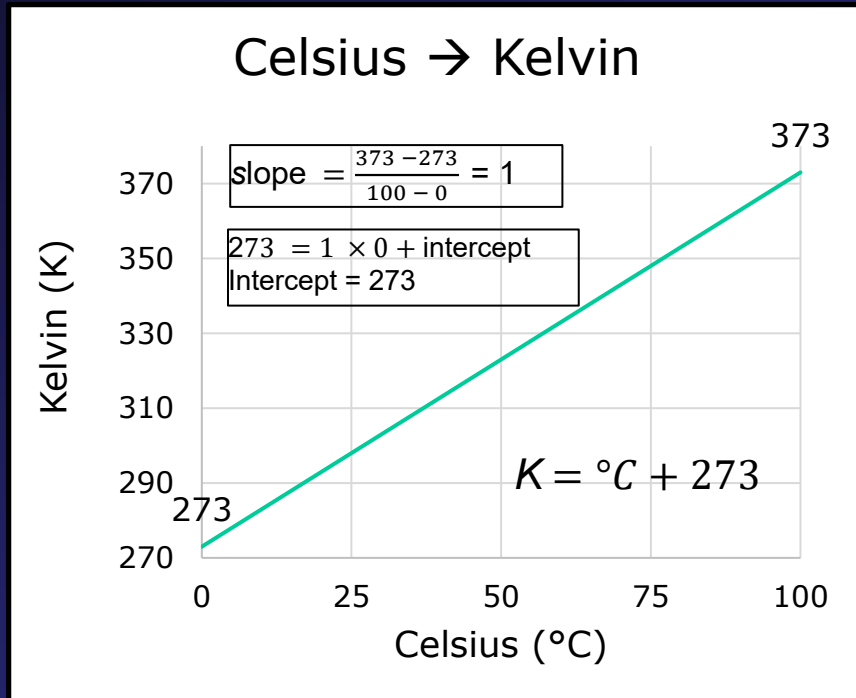
Three scales (measurement types) are used to get a value of temperature

Scale	Water Property	
	freezing point	boiling point
Fahrenheit	32° F	212° F
Celsius (centigrade)	0° C	100° C
Kelvin	273 K	373 K

Thermometers usually involve expansion or contraction of calibrated sealed tube of mercury (Hg) or dyed-alcohol

Converting Temperature

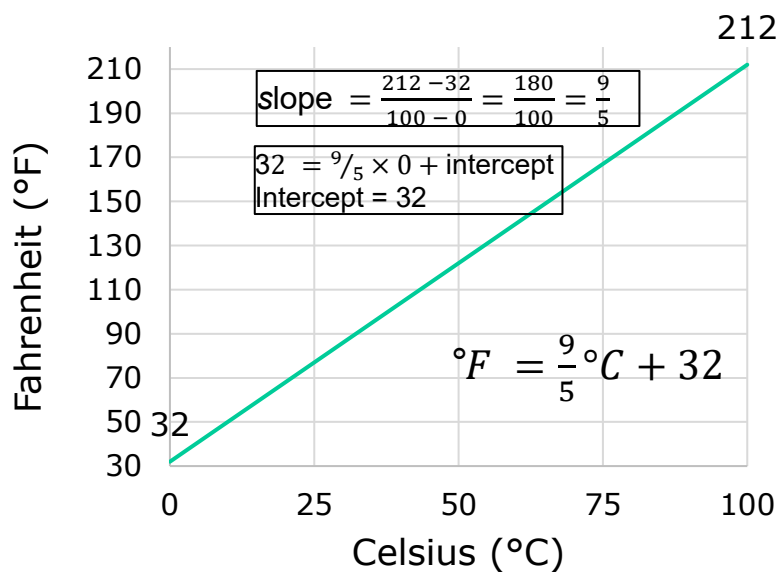
Conversions apply basic algebraic principles of sloped lines



Converting Temperature

Fahrenheit not used in chemistry, but be prepared to convert

Celsius → Fahrenheit



Fahrenheit → Celsius

